



NEUTRON & X-RAY OPTICS

Shaping, focusing, and timing the beam

Mirrors, lenses, crystals, multilayers and choppers in McStas & McXtrace

McStas — neutrons

McXtrace — X-rays



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PaNRAID School · supported by PEPR DIADEM

What we'll cover

Everything between source and sample that reshapes the beam in space, direction, energy or time.

1 Five ways to shape a beam

A quick map of the optics families we'll cover

3 Grazing-incidence reflection

The shared physics behind guides and mirrors

5 Refractive lenses (CRLs)

Why X-ray and neutron lenses must be stacked

7 Multilayers

One interference effect, two very different jobs

2 Spatial & directional shaping

Slits, apertures and collimators

4 Guides & mirrors

Channelling neutrons vs. focusing X-rays

6 Crystal monochromators

Bragg's law, mosaicity, and geometry

8 Temporal shaping

Choppers, velocity selectors, and why neutrons are special



Five ways to shape a beam

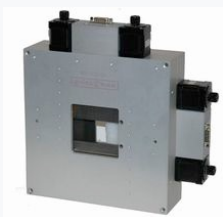
Every optical component in either code falls into one of these families.

Spatial	Reflective	Refractive	Diffractive	Temporal
1 Pinholes, slits, apertures, collimators — define where the beam is allowed to be.	2 Mirrors & guides — steer or focus at grazing incidence.	3 Compound lenses (CRLs) — bend the beam like an ordinary lens, just barely.	4 Crystals & multilayers — select wavelength/energy via interference.	5 Choppers & selectors — shape the beam in time, not space.

We'll move roughly in this order — spatial first, temporal last — flagging where neutrons and X-rays share a component and where they don't.

Spatial & directional shaping

Spatial: pinholes, slits, apertures

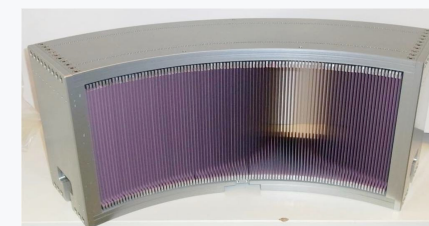


- A static or adjustable “hole” in absorbing material
- Defines the beam's spatial footprint downstream
- Same component idea in both codes: Slit

Directional: collimators



- Two or more slits (or a channel array) define an accepted divergence
- Absorbing walls create parallel “channels” through the collimator
- Usually static, but rotating radial collimators exist — e.g. to reject sample-environment scattering



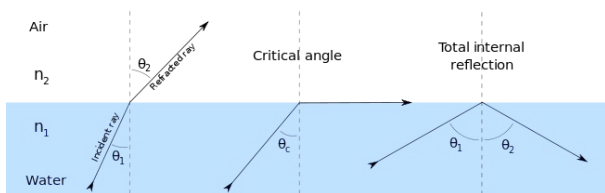
Fully shared: these are geometric, energy-independent devices — the same component names and logic apply whether the beam is neutrons or photons.



Grazing-incidence reflection

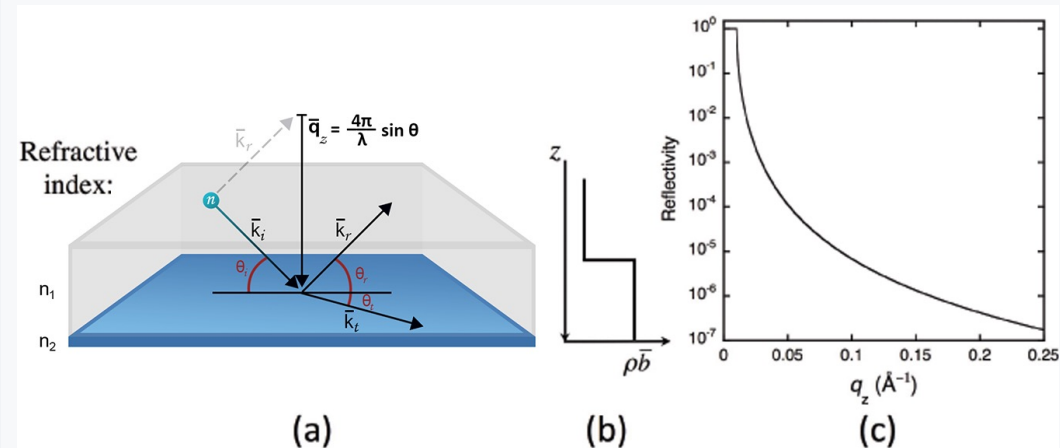
- For both neutrons and X-rays, matter's refractive index is just below 1: $n = 1 - \delta$ (+ $i\beta$ for absorption).
- Going from vacuum ($n=1$) into a medium with $n<1$ mirrors the classic total-internal-reflection picture — here it's called total external reflection.
- Below a small critical angle θ_c , a ray hitting a smooth, flat surface reflects almost perfectly — no coating chemistry required, just the right angle.
- This single effect underlies both neutron guides and X-ray mirrors — what differs is how it's exploited.

Visible light – water/air interface:
Total **internal** reflection



Neutrons and X-rays – vacuum/matter interface:
Total **external** reflection

Below θ_c : total reflection

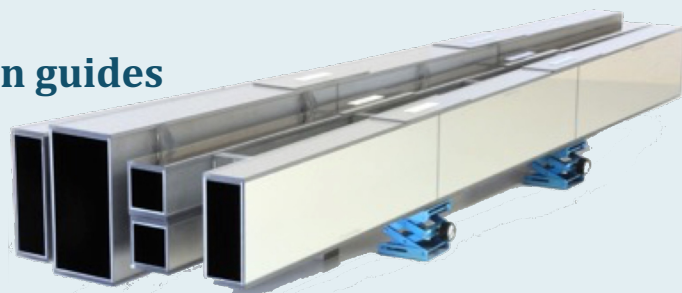


θ_c is tiny — typically a fraction of a degree — for both probes, which is why guides and mirrors are long, shallow, and need to be extremely flat/smooth.

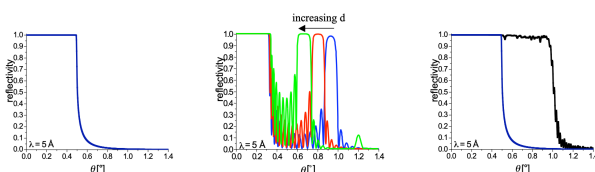
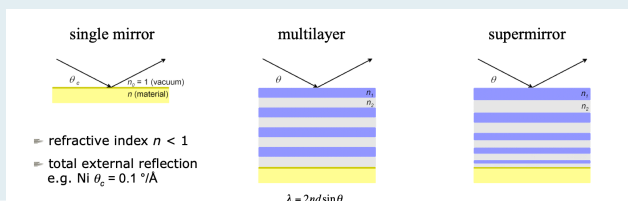
Guides & mirrors

Same critical-angle physics, but neutron beamlines and X-ray beamlines put it to work differently.

McStas · neutron guides



- Guide, Guide_channeled, Guide_gravity: long supermirror-coated channels
- Many shallow bounces over metres — preserves intensity over long transport distances
- Coating quality set by an “m-value” — multiples of the natural-Ni critical angle



McXtrace · X-ray mirrors

- Mirror_curved, TwinKB_ML: one or two large, single-bounce focusing mirrors
- Typically one bounce per axis (e.g. a Kirkpatrick-Baez pair) rather than a many-bounce channel
- Reflectivity data files (measured or computed) describe the coating's angle/energy response

Why the difference? Neutron sources are weak per-neutron, so long guides trade length for preserved flux; X-ray sources are so bright that a couple of well-figured mirrors are enough to focus or steer the beam.

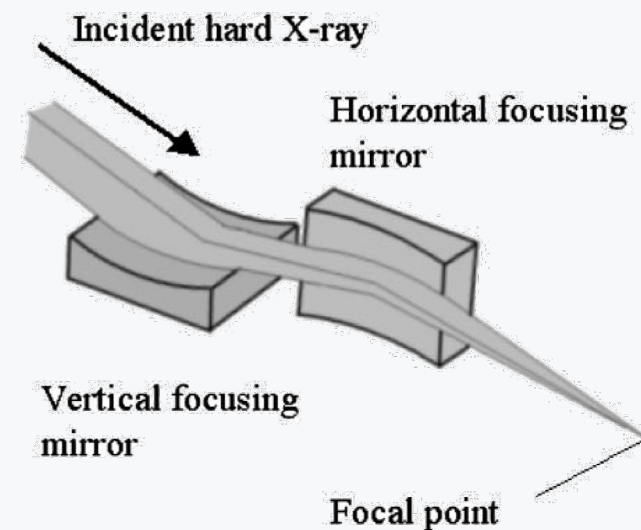
KB mirrors, in practice

A Kirkpatrick-Baez (KB) pair: two perpendicular curved mirrors, one per focusing axis.

- First (meridional) mirror focuses one axis; a glancing angle θ redirects the beam by 2θ .
- An Arm component re-establishes the optical axis after each reflection — it does nothing physically, it just carries a rotated coordinate frame forward.
- The second mirror sits on an Arm rotated 90° around the beam axis, so it focuses the perpendicular direction.
- Focal lengths follow directly from the glancing angle and radius of curvature R :

$$\begin{aligned} f_m &= R \cdot \sin(\theta/2) && \text{(meridional / 1st mirror)} \\ f_s &= R/2 \cdot \sin(\theta) && \text{(sagittal / 2nd mirror)} \end{aligned}$$

source $\rightarrow$ mirror 1 $\rightarrow$ mirror 2 $\rightarrow$ focus



ϑ is typically only a few mrad — chosen from the material's critical angle at the working energy.

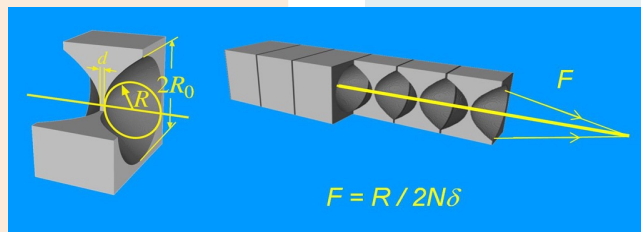
Refractive lenses (CRLs)

Both codes offer compound refractive lenses — and both need them for the same reason.

X-ray

$$\delta_{\text{Xray}} = f(Z, \lambda)$$

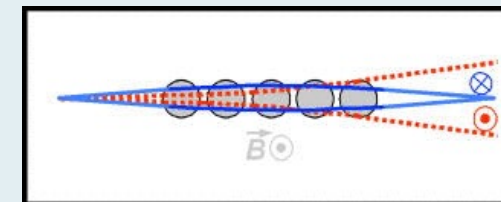
- Grows with electron density — higher atomic number Z means more refraction, but also more absorption
- Low- Z , low-absorption materials preferred: Be, Al, polymers



Neutron

$$n = 1 - \delta_{\text{neutron}} = f(b, B)$$

- Governed by coherent scattering length b , plus a magnetic term B
- The magnetic (spin-dependent) lensing effect is small in practice — hence ordinary CRLs are still the practical choice



Why “compound”: in both regimes the index is barely below 1, so a single lens bends the beam almost imperceptibly. Stacking many (parabolic) lenses in series — a compound refractive lens — adds up to a usable focal length.

A CRL beamline, worked example

Stack	Material	R (tip)	Lenses	Diameter	Thickness (tip)
Stack 1	Be	200 μm	16	1 mm	50 μm
Stack 2	Al	200 μm	10	1 mm	20 μm

Setup: Source_gaussian at $E_0=23.32$ keV ($dE=1$ keV), CRL stack 31.5 m downstream, focused to a spot 10 m further on. A 2D + energy monitor at the stack checks the incoming footprint and spectrum; another monitor is scanned in z to find the true focus.

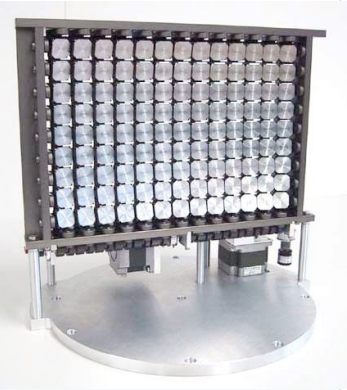
- Directional sampling (focus_xw, focus_yh, dist on the source) concentrates statistics on the aperture that matters, without biasing intensity.
- Used together with a slit, the CRL's achromatic focusing can double as a crude, high-throughput monochromator.
- mxdisplay ("Trace" mode) lets you visually follow rays through the lens stack — useful for sanity-checking alignment.
- Swapping Be $\leftrightarrow$ Al lets you compare absorption vs. refraction trade-offs directly.



Crystal monochromators: Bragg's law

`n λ = 2 d sin(θ)` — exactly the same equation, in both codes

	X-ray	Neutron
Typical crystal	Perfect single crystal: Si, Ge, C	Mosaic crystal: pyrolytic graphite, deformed Cu, Ge
Mosaicity	Very low	Often high (multi-crystal, focusing arrays)
Geometry	Laue (transmission) or Bragg (reflection)	Almost always Bragg (reflection)
Resulting beam	Narrow, low divergence and $\frac{\delta\lambda}{\lambda} \approx 0.1\%$	Medium divergence and $\frac{\delta\lambda}{\lambda} \approx 3\%$



In both codes: Bragg_crystal (McXtrace) and Monochromator_flat / Monochromator_curved / Single_crystal (McStas) all rotate to satisfy Bragg's law — the rotation angle is a natural scan parameter for a rocking curve.

Bent crystals & double-crystal monochromators

Bending a crystal

- A radius of curvature on the crystal (Bragg_crystal_bent) is a simple way to model a heat-bump — thermal distortion from absorbed beam power.
- Compare a Divergence_monitor before/after bending to see the extra divergence introduced.
- A rocking curve (scanning θ) shows the peak shape change directly.

Double-crystal monochromators (DCM)

- A second crystal, rotated by the negative of the first angle, brings the beam back parallel to the incoming direction
- Arms track the beam axis between crystals; the pair can share a single rotation stage, as in a real DCM
- Laue_crystal_BC swaps reflection for transmission geometry — useful for higher-energy or higher-heat-load beamlines

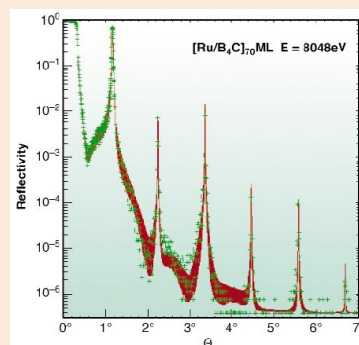
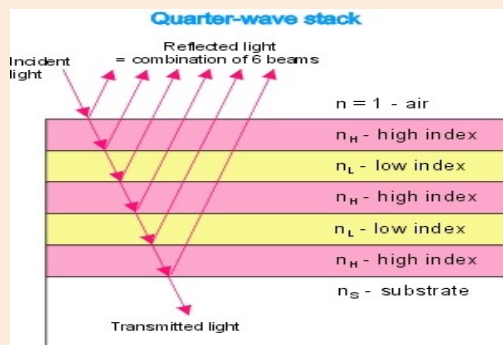
```
// crystal 2, mirrored relative to crystal 1's Arm
COMPONENT xtal2 = Bragg_crystal(...)
AT (0,0,0) RELATIVE arm_after_xtal1
ROTATED (-MONO_ANGLE, 0, 0) RELATIVE arm_after_xtal1
```

Multilayers: one concept, two jobs

A stack of thin alternating layers reflects by interference — like Bragg's law, but for a synthetic superperiod.

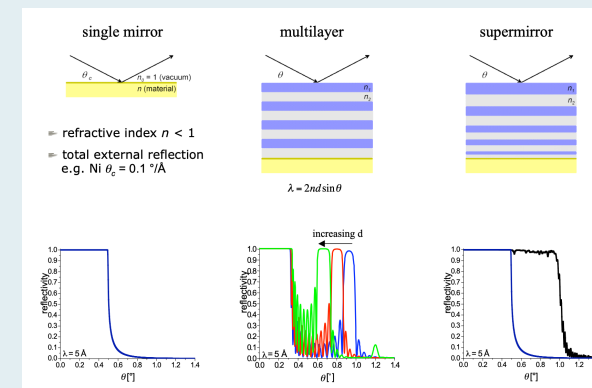
X-ray: a broadband monochromator

- Used as a monochromator with a much broader bandpass — and higher flux — than a perfect crystal
- Pick one of the first “side peaks” on the reflectivity curve
- Curved multilayers combine monochromation with focusing



Neutron: a “supermirror” coating

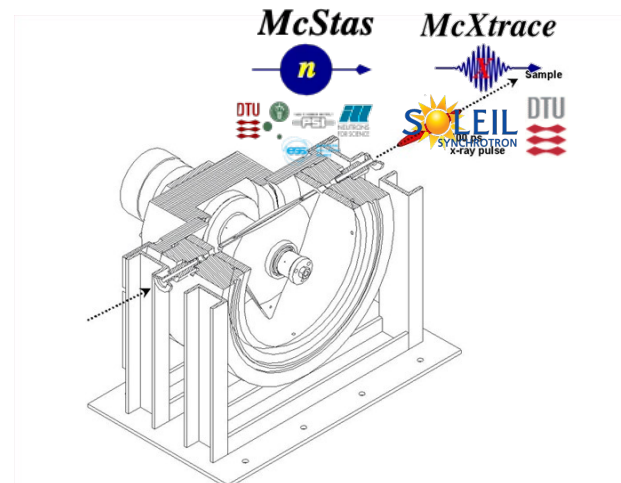
- Stacked Bragg-like reflections extend the total-external-reflection range well beyond a single critical angle
- Ideally a wide, flat reflectivity vs. Q
- Used to coat neutron guides — effectively an “optical fibre” for neutrons



Same underlying phase-interference physics in both codes — only the intended job (monochromate vs. transport) differs.

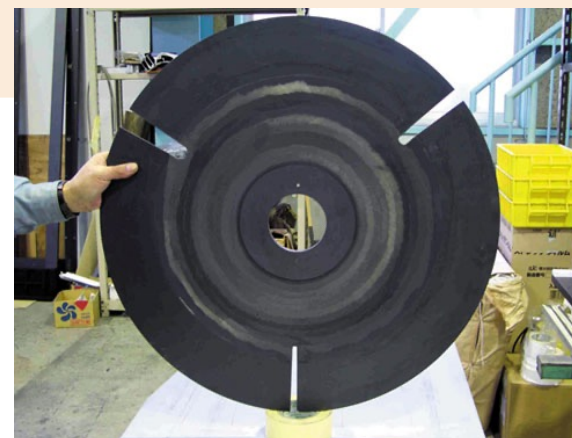
Temporal shaping: choppers

- A rotating absorbing disk with one or more slits cuts a continuous beam into pulses in time.
- Both X-ray and neutron beamlines use a disk-chopper-like component to define or measure time structure — e.g. for time-of-flight (TOF) measurements.
- The physical picture is identical: a slit sweeps across the beam at a set frequency, and only rays arriving within the open window get through.



Where it stops being shared: only neutrons are slow enough that a mechanical rotor can also act as an energy filter. An X-ray chopper shapes time only — it can't pick out an energy band the way a neutron velocity selector or Fermi chopper can.

Up next: two neutron-only rotating devices that filter on speed, not just on time.



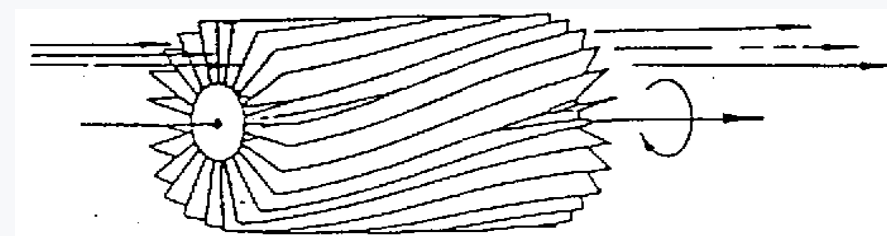
Velocity selectors

A rotating drum of twisted, absorbing blades — only a narrow velocity (wavelength) band makes it straight through.

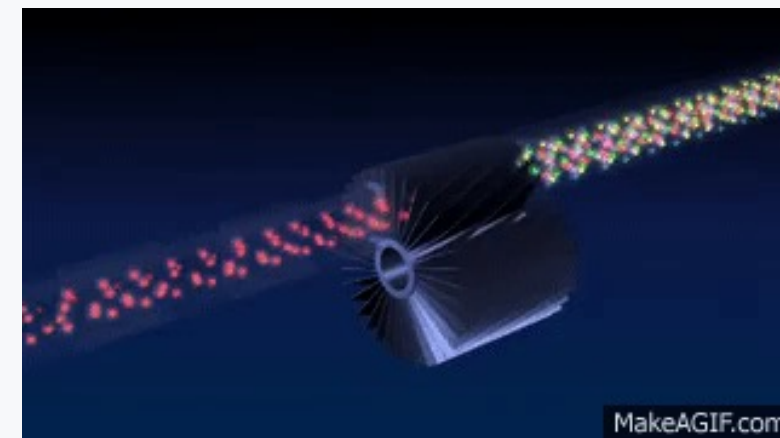


Parameter	Meaning
xwidth, yheight [m]	entry aperture
zdepth [m]	housing length
length, d [m]	blade length / thickness
alpha [deg]	blade twisting angle
radius [m]	rotation axis → aperture-centre distance
nu [Hz]	rotation speed (counterclockwise)
nslit	number of blades

twisted absorbing blades



broad monochromatization
 $\delta\lambda/\lambda \approx 10\%$



MakeAGIF.com

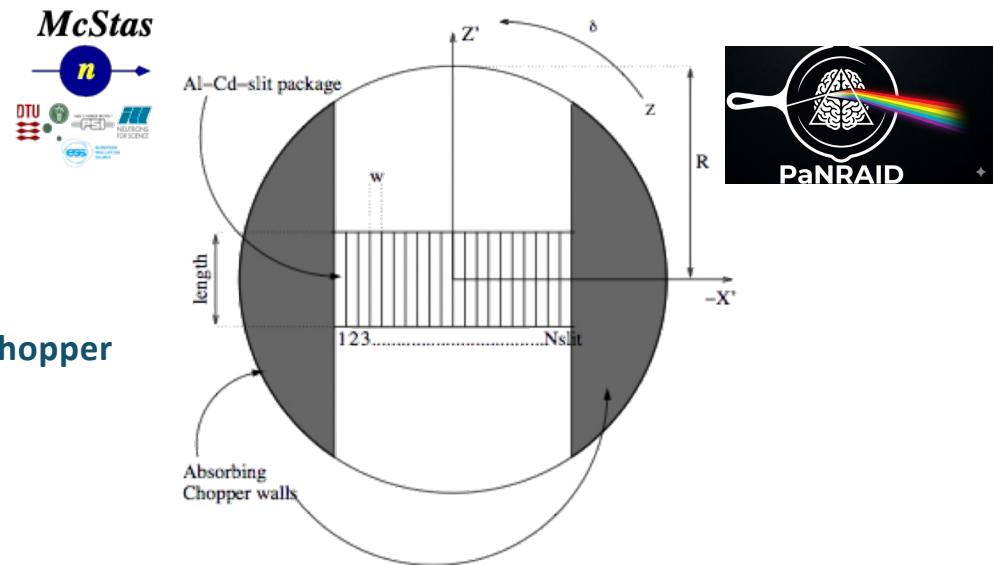
Disk & Fermi choppers

DiskChopper

Parameter	Meaning
<code>nu, theta_0</code>	frequency; angular slit width
<code>radius, yheight, xwidth</code>	disk geometry / slit opening
<code>jitter, delay, phase</code>	timing: random jitter, fixed delay, or angular phase
<code>isfirst, npulse</code>	marks the first chopper; pulses to generate if so

Both are McStas-only: there's no X-ray analogue — the mechanical rotation speeds needed would be absurd at the speed of light. X-ray timing instead comes from the accelerator's bunch structure itself.

Fermi chopper



- A small drum of curved slit packages, spinning fast, sitting directly in the beam — a “rotating collimator”.
- Only neutrons with the right speed can cross the curved channel before it rotates shut — so it acts as a combined fast time-gate and coarse energy filter.
- Used right next to the sample, where a disk chopper's timing alone is no longer selective enough.



One optics toolkit, two beams

What's shared

- Spatial shaping: slits & collimators, same logic either way
- Grazing-incidence total external reflection ($n < 1$, critical angle)
- Bragg's law, and the idea of rotating a crystal to tune it
- Refractive-index-just-below-1 logic behind compound lenses
- Multilayer interference physics
- The chopper concept for defining time structure

What differs

- Many-bounce guides (neutron) vs. one/two-bounce focusing mirrors (X-ray)
- Crystal choice: near-perfect low-mosaicity vs. mosaic/multi-crystal
- Multilayer role: X-ray monochromator vs. neutron guide coating
- Velocity selectors & Fermi choppers exist only for neutrons
- X-ray timing comes from the accelerator, not a mechanical rotor

Where to go deeper

McStas

`mcdoc optics / mcdoc -c`

Guide, Guide_channeled, Monochromator_flat/curved, DiskChopper, FermiChopper, V_selector

mcstas.org

McXtrace

`mxdoc optics / mxdoc -c`

Practical exercises: CRLs, KB mirrors, monochromators — pick two of the three

mcxtrace.org

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